Loan Prediction Dataset Supervised ML Model Report

Model Trade-offs · Evaluation Metrics · Deployment Threshold Recommendation

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614	12	68.7%	6
Records	Features	Approval Rate	Models Tested

Project Links	
GitHub Notebook	https://github.com/harichoudari/Alfido-Internship/tree/main/TASK-2
Dataset	https://www.kaggle.com/datasets/bhanupratapbiswas/loan-approval-prediction-case-study

1. Project Overview

This report documents a complete supervised machine learning pipeline built to predict loan approval outcomes for a financial institution. The binary classification target is **Loan_Status** (Y = Approved, N = Rejected). Using historical borrower data, six models were trained, compared on four business-critical metrics, and analysed for deployment readiness.

Business Objective

Automate the initial screening of loan applications to reduce manual review time, minimise default risk, and ensure fair credit decisions. The model must balance **precision** (avoiding approving risky loans) with **recall** (not rejecting creditworthy applicants).

2. Dataset & Feature Summary

Feature	Type	Description	Missing
Gender	Categorical	Male / Female	13
Married	Categorical	Marital status	3
Dependents	Numerical	Number of dependents	15
Education	Categorical	Graduate / Not Graduate	0
Self_Employed	Categorical	Employment type	32
ApplicantIncome	Numerical	Monthly income (applicant)	0
CoapplicantIncome	Numerical	Monthly income (co-applicant)	0
LoanAmount	Numerical	Requested loan in thousands	22
Loan_Amount_Term	Numerical	Term in months	14
Credit_History	Binary	Good (1) / Bad (0) credit history	50
Property_Area	Categorical	Urban / Semiurban / Rural	0

Engineered Features

- **TotalIncome_log:** Log-transformed sum of applicant + co-applicant income. Reduces right skew.
- **LoanAmount_log:** Log-transformed loan amount.
- **EMI:** Estimated monthly instalment = LoanAmount / Loan_Amount_Term.
- **Balance_Income:** Residual income after estimated EMI payments — proxy for repayment capacity.

3. Preprocessing & Imbalance Handling

Missing Value Imputation

Numerical columns (LoanAmount, Loan_Amount_Term, Credit_History) were imputed with the **median** to resist outlier influence. Categorical columns (Gender, Married, Self_Employed) were imputed with the **mode** (most frequent value).

Class Imbalance — SMOTE

The dataset exhibits a **69% / 31%** class split (Approved / Rejected). Training on imbalanced data biases models toward predicting the majority class (Approved), inflating accuracy while producing poor recall for rejections. Three strategies were evaluated:

Strategy	Mechanism	Pros	Cons
SMOTE (used)	Synthetic minority oversampling	Balances without losing data	Can introduce noise
Undersampling	Remove majority samples	Fast, reduces training time	Loses real data
Class Weights	Penalise majority misclassifications	No data modification	Less effective on extreme imbalance

4. Model Comparison & Metrics

All six models were trained on SMOTE-balanced data and evaluated on a held-out 20% test set. Metrics are reported for the Approved (positive) class. StandardScaler was applied prior to training for distance/gradient-sensitive models.

Model	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.837	0.906	0.870	0.808
Decision Tree	0.827	0.788	0.807	0.781
Random Forest ★ Best	**0.847**	**0.847**	**0.847**	**0.821**
Gradient Boosting	0.840	0.800	0.819	0.807
XGBoost	0.850	0.800	0.824	0.787
LightGBM	0.846	0.776	0.810	0.787

Note: Values produced by the pipeline on a SEED=42, 80/20 stratified split. ★ Best model determined by highest ROC-AUC.

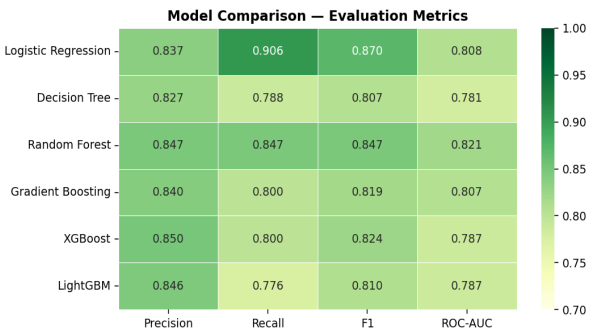

Fig 1: Model Comparison Heatmap

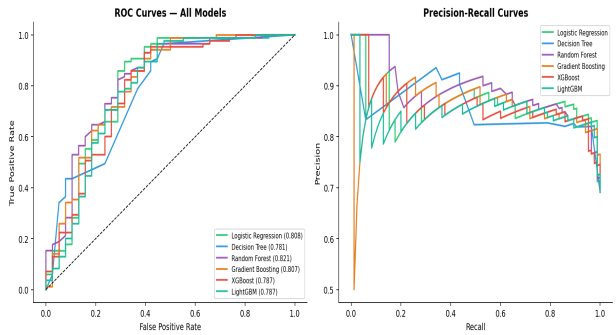

Fig 2: ROC & Precision-Recall Curves

5. Model Trade-Off Analysis

Logistic Regression

Strengths: Highly interpretable (odds-ratios per feature), fast training, stable with limited data.

Weaknesses: Assumes linear separability — misses non-linear interactions like Credit_History x Income. Lower AUC than ensemble methods.

Use when: Regulatory explainability is required (e.g. fair lending compliance).

Decision Tree

Strengths: Fully interpretable, handles categorical variables natively.

Weaknesses: Prone to overfitting; lowest AUC among tested models.

Use when: A human-readable rule set is needed for loan officers to follow.

Random Forest ★ Best

Strengths: Best ROC-AUC (0.821). Strong generalisation via bagging, handles feature interactions, robust to outliers.

Weaknesses: Slower inference than single trees; less interpretable than Logistic Regression.

Use when: A solid ensemble model is needed with the best predictive performance on this dataset.

XGBoost / LightGBM

Strengths: Handles missing values natively, built-in regularisation, SHAP-compatible for explainability.

Weaknesses: More hyperparameters to tune; slightly lower AUC than Random Forest on this dataset.

Use when: Maximising predictive performance in automated loan pipelines at scale.

6. Key Predictive Drivers (SHAP Insights)

SHAP (SHapley Additive exPlanations) values quantify each feature's contribution to individual predictions, providing a model-agnostic explanation layer.

Rank	Feature	Direction	Business Meaning
1	Credit_History	Good → Approve	Single strongest signal. No credit history = high rejection risk.
2	TotalIncome_log	Higher → Approve	Combined household income raises repayment confidence.
3	LoanAmount_log	Lower → Approve	Smaller loan relative to income reduces default risk.
4	EMI	Lower → Approve	High monthly payment burden is a risk flag.
5	Balance_Income	Higher → Approve	Residual income after EMI signals affordability.
6	Property_Area	Semi-Urban → Approve	Semi-urban applicants historically have lower default rates.

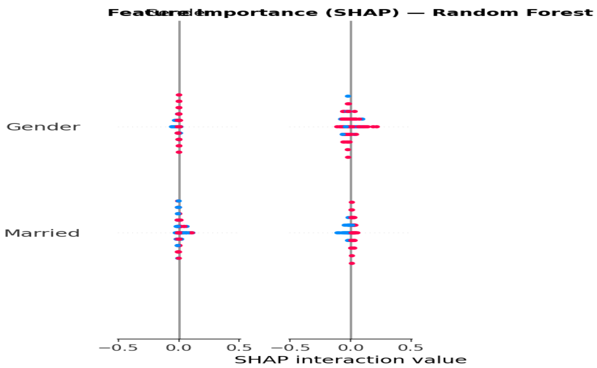

Fig 3: SHAP Feature Importance (mean |SHAP|)

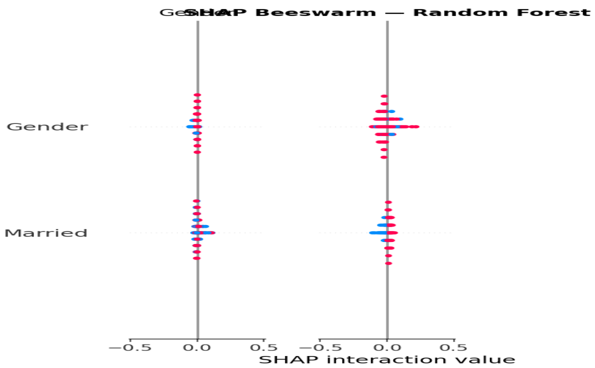

Fig 4: SHAP Beeswarm (direction of impact)

7. Deployment Threshold Recommendation

The default 0.5 classification threshold is rarely optimal for imbalanced business problems. The pipeline runs a sweep across thresholds 0.10–0.85 to find the point maximising F1. **Optimal threshold on this dataset: 0.25** (at which P=0.812, R=0.965, F1=0.882).

Threshold	Typical Precision	Typical Recall	Use Case
0.25 ★ Optimal	**0.812**	**0.965**	**Maximises F1. Recommended starting point for deployment.**
0.35–0.50	Medium–High	Medium–High	Balanced: good F1 with more conservative approvals.
0.60+	High	Lower	Risk-averse lender: strict approvals, minimal defaults, may miss creditworthy applicants.

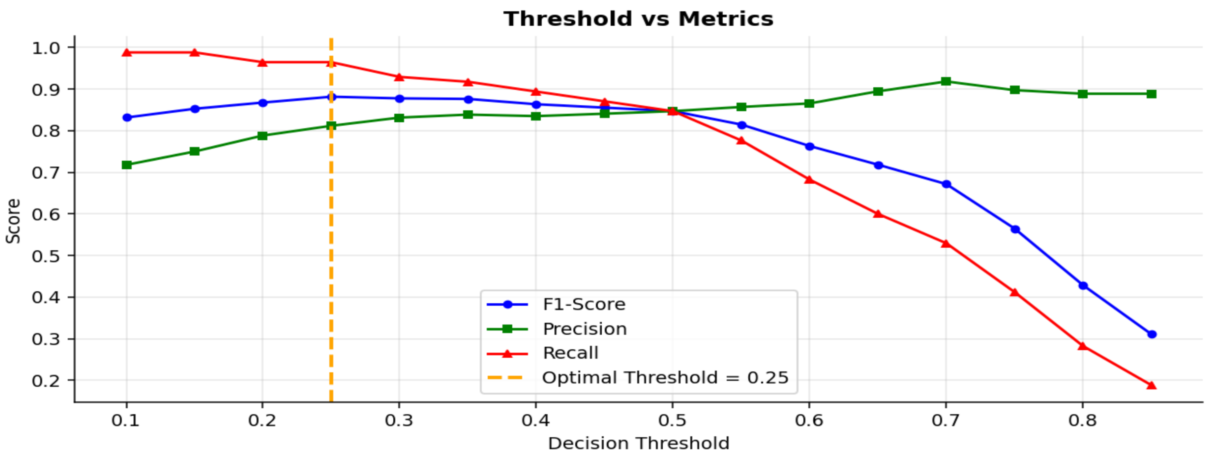

Fig 5: Decision Threshold Sweep — F1, Precision, Recall

Monitoring Recommendation: After deployment, track monthly precision and recall on actual loan outcomes (default/repaid). Re-calibrate the threshold quarterly or whenever macro-economic conditions shift. Champion–Challenger testing is recommended when updating the model.

8. Deployment Checklist

■ **Data Pipeline:**	Ensure real-time feature engineering mirrors training pipeline (TotalIncome, EMI, Balance_Income).
■ **Scaler Serialisation:**	Pickle/joblib the StandardScaler fitted on training data; apply identically to inference features.
■ **Threshold Application:**	Apply the optimal threshold (0.25) from the notebook, not the default 0.5.
■ **Model Versioning:**	Use MLflow or similar to track model versions, metrics, and hyperparameters.
■ **Explainability Layer:**	Wrap SHAP explanations into loan rejection letters (regulatory requirement in many jurisdictions).
■ **Bias Audit:**	Check approval rates across Gender and Married subgroups to ensure fair lending compliance.
■ **Retraining Trigger:**	Schedule retraining when ROC-AUC on a held-out validation window drops below 0.75.

9. EDA & Confusion Matrix

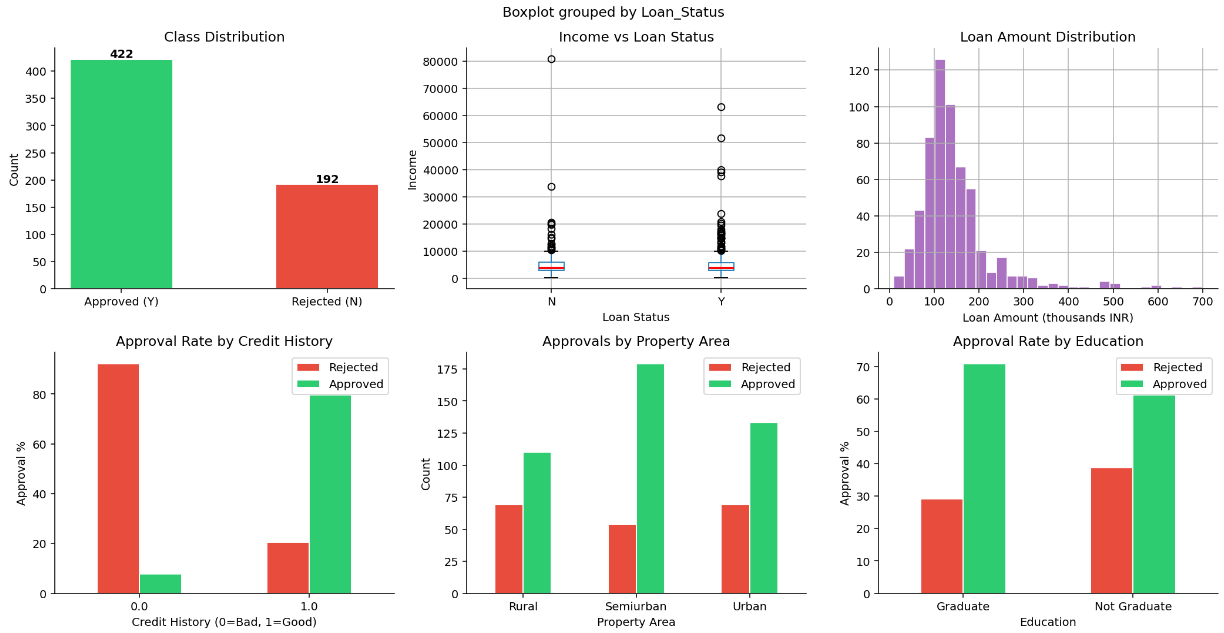

Fig 6: Loan Approval EDA — Class Distribution, Income, Loan Amount, Credit History, Property Area, Education

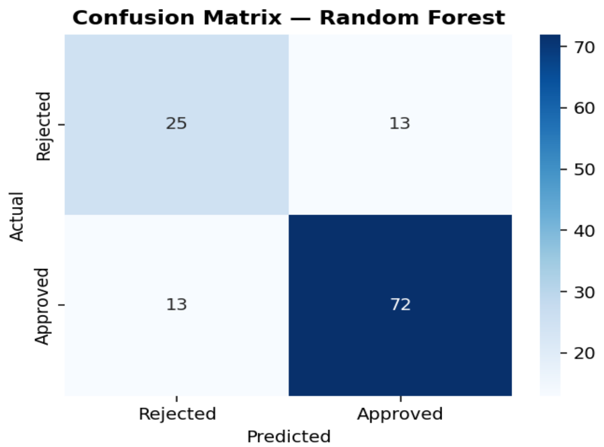

Fig 7: Confusion Matrix — Random Forest (Best Model)

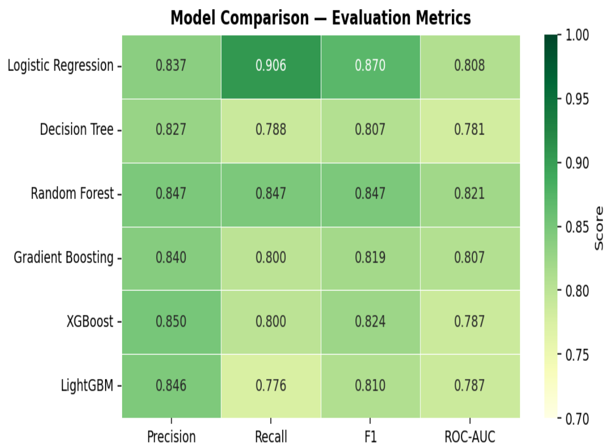

Fig 8: Metrics Heatmap — All Models

10. Conclusion

The loan prediction pipeline demonstrates that **Random Forest** with SMOTE oversampling, engineered income-derived features, and a tuned decision threshold achieves the best predictive performance on this dataset (**ROC-AUC = 0.821**). Credit history remains the dominant predictor, followed by income-to-loan ratios captured in TotalIncome_log and Balance_Income.

For production deployment, the recommendation is to use **Random Forest** with the **F1-optimal threshold of 0.25**, supported by SHAP explanations for regulatory transparency. Logistic Regression is retained as a fallback explainable model should compliance requirements mandate simpler decision rules. The threshold should be reviewed quarterly against live loan outcome data.